

Telecom Customer Churn Dataset Analysis and Prediction in R

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Data Cleaning

First, we install and load necessary libraries that we will use.

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(tidyverse)

## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ forcats 1.0.0      ✓ readr 2.1.4
## ✓ ggplot2 3.5.1      ✓ stringr 1.5.0
## ✓ lubridate 1.9.3    ✓ tibble 3.2.1
## ✓ purrr 1.0.2        ✓ tidyr 1.3.0

## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag() masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
## conflicts to become errors

library(caret)

## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##   lift

library(ggplot2)
library(GGally)

## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg ggplot2
```

We install and load the readxl package to read the excel file

```
library(readxl)
Churn_Dataset <- read_excel("F:/Datasets/Churn-Dataset.xlsx")
```

Now, we should look at the data and check the dimension of th churn_data

```
churn_data <- Churn_Dataset
head(churn_data)

## # A tibble: 6 × 23
##   customerID gender SeniorCitizen Partner Dependents tenure PhoneService
##   <chr>      <chr>      <dbl> <chr>    <chr>      <dbl> <chr>
## 1 7590-VHVEG Female          0 Yes      No          1 No
## 2 5575-GNVDE Male          0 No       No         34 Yes
## 3 3668-QPYBK Male          0 No       No          2 Yes
## 4 7795-CFOCW Male          0 No       No         45 No
## 5 9237-HQITU Female        0 No       No          2 Yes
## 6 9305-CDSKC Female        0 No       No          8 Yes
## # [i] 16 more variables: MultipleLines <chr>, InternetService <chr>,
## #   OnlineSecurity <chr>, OnlineBackup <chr>, DeviceProtection <chr>,
## #   TechSupport <chr>, StreamingTV <chr>, StreamingMovies <chr>,
## #   Contract <chr>, PaperlessBilling <chr>, PaymentMethod <chr>,
## #   MonthlyCharges <dbl>, TotalCharges <dbl>, numAdminTickets <dbl>,
## #   numTechTickets <dbl>, Churn <chr>

dim(churn_data)

## [1] 7043  23
```

We summarize our data

```
summary(churn_data)

##   customerID      gender      SeniorCitizen      Partner
## Length:7043      Length:7043      Min.   :0.0000      Length:7043
## Class :character  Class :character  1st Qu.:0.0000      Class :character
## Mode  :character  Mode  :character  Median :0.0000      Mode  :character
##                                     Mean   :0.1621
##                                     3rd Qu.:0.0000
##                                     Max.   :1.0000
##
##   Dependents      tenure      PhoneService      MultipleLines
## Length:7043      Min.   : 0.00      Length:7043      Length:7043
## Class :character  1st Qu.: 9.00      Class :character  Class :character
## Mode  :character  Median :29.00      Mode  :character  Mode  :character
##                                     Mean   :32.37
##                                     3rd Qu.:55.00
##                                     Max.   :72.00
##
##   InternetService      OnlineSecurity      OnlineBackup      DeviceProtection
## Length:7043      Length:7043      Length:7043      Length:7043
## Class :character  Class :character  Class :character  Class :character
```

```
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## TechSupport StreamingTV StreamingMovies Contract
## Length:7043 Length:7043 Length:7043 Length:7043
## Class :character Class :character Class :character Class :character
## Mode :character Mode :character Mode :character Mode :character
##
##
##
## PaperlessBilling PaymentMethod MonthlyCharges TotalCharges
## Length:7043 Length:7043 Min. : 18.25 Min. : 18.8
## Class :character Class :character 1st Qu.: 35.50 1st Qu.: 401.4
## Mode :character Mode :character Median : 70.35 Median :1397.5
## Mean : 64.76 Mean :2283.3
## 3rd Qu.: 89.85 3rd Qu.:3794.7
## Max. :118.75 Max. :8684.8
## NA's :11
## numAdminTickets numTechTickets Churn
## Min. :0.0000 Min. :0.0000 Length:7043
## 1st Qu.:0.0000 1st Qu.:0.0000 Class :character
## Median :0.0000 Median :0.0000 Mode :character
## Mean :0.5157 Mean :0.4196
## 3rd Qu.:0.0000 3rd Qu.:0.0000
## Max. :5.0000 Max. :9.0000
##
```

We can skim the churn_data with little more details

```
skimr::skim(churn_data)
```

Data summary

```
Name churn_data
Number of rows 7043
Number of columns 23
```

Column type frequency:

```
character 17
numeric 6
```

```
Group variables None
```

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
customerID	0	1	10	10	0	7043	0
gender	0	1	4	6	0	2	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Partner	0	1	2	3	0	2	0
Dependents	0	1	2	3	0	2	0
PhoneService	0	1	2	3	0	2	0
MultipleLines	0	1	2	16	0	3	0
InternetService	0	1	2	11	0	3	0
OnlineSecurity	0	1	2	19	0	3	0
OnlineBackup	0	1	2	19	0	3	0
DeviceProtection	0	1	2	19	0	3	0
TechSupport	0	1	2	19	0	3	0
StreamingTV	0	1	2	19	0	3	0
StreamingMovies	0	1	2	19	0	3	0
Contract	0	1	8	14	0	3	0
PaperlessBilling	0	1	2	3	0	2	0
PaymentMethod	0	1	12	25	0	4	0
Churn	0	1	2	3	0	2	0

Variable type: numeric

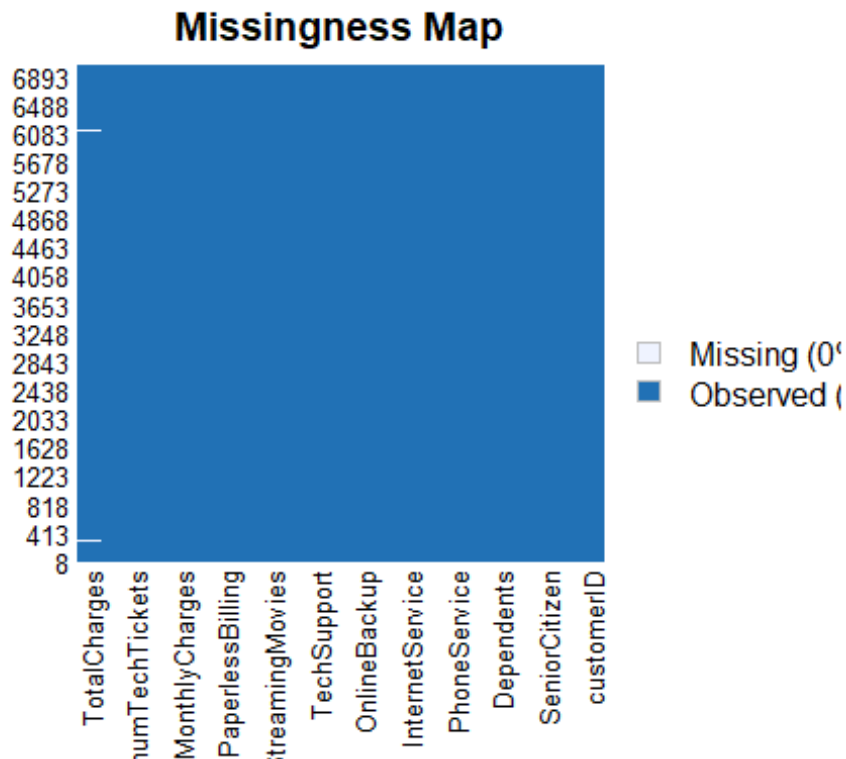
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
SeniorCitizen	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
tenure	0	1	32.37	24.56	0.00	9.00	29.00	55.00	72.00	
MonthlyCharges	0	1	64.76	30.09	18.25	35.50	70.35	89.85	118.75	
TotalCharges	11	1	2283.30	2266.77	18.80	401.45	1397.47	3794.74	8684.80	
numAdminTickets	0	1	0.52	1.28	0.00	0.00	0.00	0.00	5.00	
numTechTickets	0	1	0.42	1.25	0.00	0.00	0.00	0.00	9.00	

Lets look at the 'missingness' map of the churn_data from Amelia package

```
# missingness map
Amelia::missmap(churn_data)

## Warning: Unknown or uninitialised column: `arguments`.
## Unknown or uninitialised column: `arguments`.

## Warning: Unknown or uninitialised column: `imputations`.
```



We will look the median of the total charges for non-missing values, and replace all missing ones with this median.

```
library(dplyr)
total_charges_median <- churn_data %>%
  summarise(median(TotalCharges, na.rm = T))
# View the result
total_charges_median

## # A tibble: 1 × 1
##   `median(TotalCharges, na.rm = T)`
##                               <dbl>
## 1                               1397.
```

Median of TotalCharges is 1397.475 . Now we replace na values to median

```
churn_df <-
  churn_data %>%
    mutate(TotalCharges =
      replace(TotalCharges, is.na(TotalCharges), total_charges_median)) %>%
    mutate(TotalCharges = unlist(TotalCharges))
skimr::skim(churn_df)
```

Data summary

Name	churn_df
Number of rows	7043
Number of columns	23

Column type frequency:

character 17
numeric 6

Group variables None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
customerID	0	1	10	10	0	7043	0
gender	0	1	4	6	0	2	0
Partner	0	1	2	3	0	2	0
Dependents	0	1	2	3	0	2	0
PhoneService	0	1	2	3	0	2	0
MultipleLines	0	1	2	16	0	3	0
InternetService	0	1	2	11	0	3	0
OnlineSecurity	0	1	2	19	0	3	0
OnlineBackup	0	1	2	19	0	3	0
DeviceProtection	0	1	2	19	0	3	0
TechSupport	0	1	2	19	0	3	0
StreamingTV	0	1	2	19	0	3	0
StreamingMovies	0	1	2	19	0	3	0
Contract	0	1	8	14	0	3	0
PaperlessBilling	0	1	2	3	0	2	0
PaymentMethod	0	1	12	25	0	4	0
Churn	0	1	2	3	0	2	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
SeniorCitizen	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
tenure	0	1	32.37	24.56	0.00	9.00	29.00	55.00	72.00	
MonthlyCharges	0	1	64.76	30.09	18.25	35.50	70.35	89.85	118.75	
TotalCharges	0	1	2281.92	2265.27	18.80	402.22	1397.47	3786.60	8684.80	
numAdminTickets	0	1	0.52	1.28	0.00	0.00	0.00	0.00	5.00	
numTechTickets	0	1	0.42	1.25	0.00	0.00	0.00	0.00	9.00	

Let's check the duplicate values of all the variables in the churn_data

```
duplicates <- duplicated(churn_df)
duplicates
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```
## [6565] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6577] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6589] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6601] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6613] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6625] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6637] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6649] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6661] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6673] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6685] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6697] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6709] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6721] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6733] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6745] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6757] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6769] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6781] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6793] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6805] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6817] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6829] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6841] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6853] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6865] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6877] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6889] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6901] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6913] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6925] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6937] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6949] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6961] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6973] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6985] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [6997] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [7009] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [7021] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [7033] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

We will convert all character variables into factor variables and glimpse the data to show the data type and values of all the variables

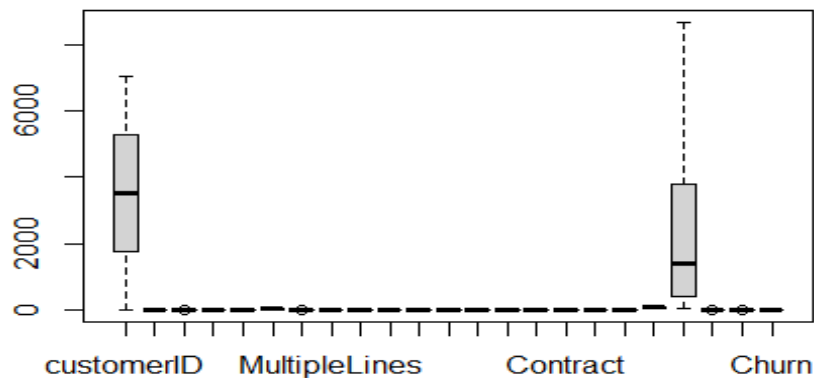
```
churn_df <- churn_df %>% mutate(across(where(is.character), as.factor))
dplyr::glimpse(churn_df)

## Rows: 7,043
## Columns: 23
## $ customerID      <fct> 7590-VHVEG, 5575-GNVDE, 3668-QPYBK, 7795-CFOCW, 9237-...
## $ gender          <fct> Female, Male, Male, Male, Female, Female, Male, Femal...
## $ SeniorCitizen    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
## $ Partner          <fct> Yes, No, No, No, No, No, No, No, No, Yes, No, Yes, No, Ye...
## $ Dependents       <fct> No, No, No, No, No, No, No, Yes, No, No, Yes, Yes, No, No...
```

```
## $ tenure      <dbl> 1, 34, 2, 45, 2, 8, 22, 10, 28, 62, 13, 16, 58, 49, 2...
## $ PhoneService <fct> No, Yes, Yes, No, Yes, Yes, Yes, No, Yes, Yes, Yes, Y...
## $ MultipleLines <fct> No phone service, No, No, No phone service, No, Yes, ...
## $ InternetService <fct> DSL, DSL, DSL, DSL, Fiber optic, Fiber optic, Fiber o...
## $ OnlineSecurity <fct> No, Yes, Yes, Yes, No, No, No, Yes, No, Yes, Yes, No ...
## $ OnlineBackup <fct> Yes, No, Yes, No, No, No, Yes, No, No, Yes, No, No in...
## $ DeviceProtection <fct> No, Yes, No, Yes, No, Yes, No, No, Yes, No, No, No in...
## $ TechSupport <fct> No, No, No, Yes, No, No, No, No, Yes, No, No, No inte...
## $ StreamingTV <fct> No, No, No, No, No, Yes, Yes, No, Yes, No, No, No int...
## $ StreamingMovies <fct> No, No, No, No, No, Yes, No, No, Yes, No, No, No inte...
## $ Contract <fct> Month-to-month, One year, Month-to-month, One year, M...
## $ PaperlessBilling <fct> Yes, No, Yes, No, Yes, Yes, Yes, No, Yes, No, Yes, No...
## $ PaymentMethod <fct> Electronic check, Mailed check, Mailed check, Bank tr...
## $ MonthlyCharges <dbl> 29.85, 56.95, 53.85, 42.30, 70.70, 99.65, 89.10, 29.7...
## $ TotalCharges <dbl> 29.85, 1889.50, 108.15, 1840.75, 151.65, 820.50, 1949...
## $ numAdminTickets <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 5, 0, 0, 0, 0,...
## $ numTechTickets <dbl> 0, 0, 0, 3, 0, 0, 0, 0, 2, 0, 0, 0, 0, 4, 0, 0, 0, 4,...
## $ Churn <fct> No, No, Yes, No, Yes, Yes, No, No, Yes, No, No, No, N...
```

We can get a clear visual of the data using a boxplot to detect the outlier

```
boxplot(churn_df)
```



We remove columnID variable from Churn dataset using select() function

```
# Removing the 'columnID' column
churn_df <- churn_df %>% select(-customerID)
```

Now we can clearly see that we don't have any unclean data using summary methods

```
summary(churn_df)
```

```

##      gender      SeniorCitizen      Partner      Dependents      tenure
## Female:3488      Min.      :0.0000      No :3641      No :4933      Min.      : 0.00
## Male  :3555      1st Qu.:0.0000      Yes:3402      Yes:2110      1st Qu.: 9.00
##                                     Median :0.0000                                     Median :29.00
##                                     Mean   :0.1621                                     Mean   :32.37
##                                     3rd Qu.:0.0000                                     3rd Qu.:55.00
##                                     Max.   :1.0000                                     Max.   :72.00
## PhoneService      MultipleLines      InternetService
## No : 682          No                   :3390      DSL                   :2421
## Yes:6361          No phone service: 682      Fiber optic:3096
##                                     Yes                   :2971      No                   :1526
##
##
##
##      OnlineSecurity      OnlineBackup
## No                       :3498      No                       :3088
## No internet service:1526      No internet service:1526
## Yes                       :2019      Yes                       :2429
##
##
##
##      DeviceProtection      TechSupport
## No                       :3095      No                       :3473
## No internet service:1526      No internet service:1526
## Yes                       :2422      Yes                       :2044
##
##
##
##      StreamingTV      StreamingMovies      Contract
## No                       :2810      No                       :2785      Month-to-month:3875
## No internet service:1526      No internet service:1526      One year       :1473
## Yes                       :2707      Yes                       :2732      Two year       :1695
##
##
##
## PaperlessBilling      PaymentMethod      MonthlyCharges
## No :2872              Bank transfer (automatic):1544      Min.      : 18.25
## Yes:4171              Credit card (automatic)  :1522      1st Qu.: 35.50
##                                     Electronic check          :2365      Median : 70.35
##                                     Mailed check             :1612      Mean   : 64.76
##                                                                   3rd Qu.: 89.85
##                                                                   Max.   :118.75
## TotalCharges      numAdminTickets      numTechTickets      Churn
## Min.      : 18.8      Min.      :0.0000      Min.      :0.0000      No :5174
## 1st Qu.: 402.2      1st Qu.:0.0000      1st Qu.:0.0000      Yes:1869
## Median :1397.5      Median :0.0000      Median :0.0000
## Mean   :2281.9      Mean   :0.5157      Mean   :0.4196
## 3rd Qu.:3786.6      3rd Qu.:0.0000      3rd Qu.:0.0000
## Max.   :8684.8      Max.   :5.0000      Max.   :9.0000

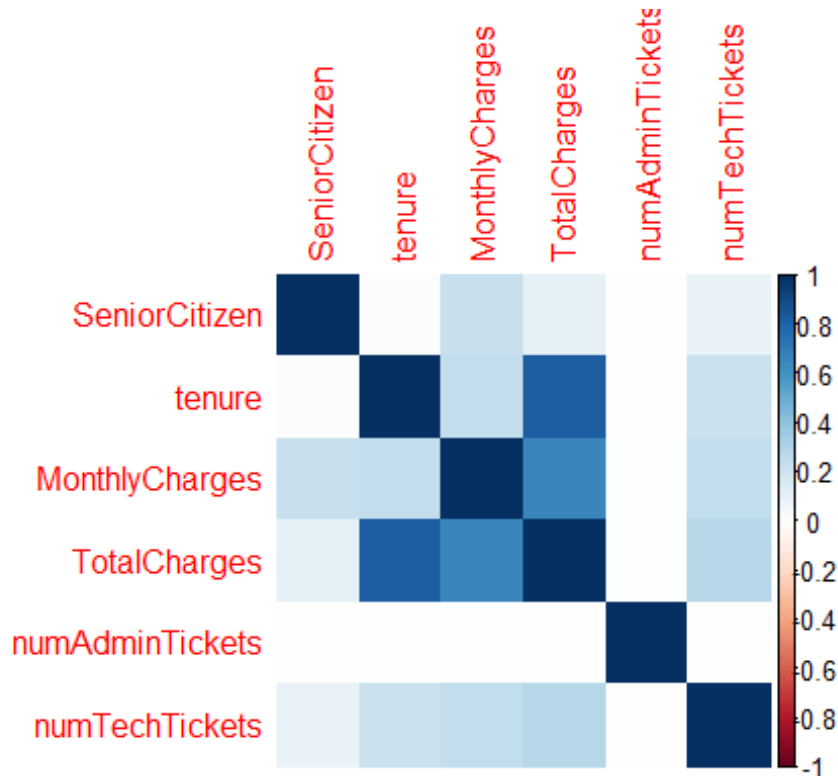
```

First we install and load the package 'corrplot' and create heatmap to find correlation between all the variables in the churn dataset

```
library("corrplot")

## corrplot 0.92 loaded

# Heatmap of Correlation matrix
cor_data <- cor(select(churn_df, c(SeniorCitizen, tenure, MonthlyCharges,
TotalCharges, numAdminTickets, numTechTickets)))
corrplot(cor_data, method="color")
```



We will create a bar graph of InternetService and PaymentMethod

```
library(gridExtra)

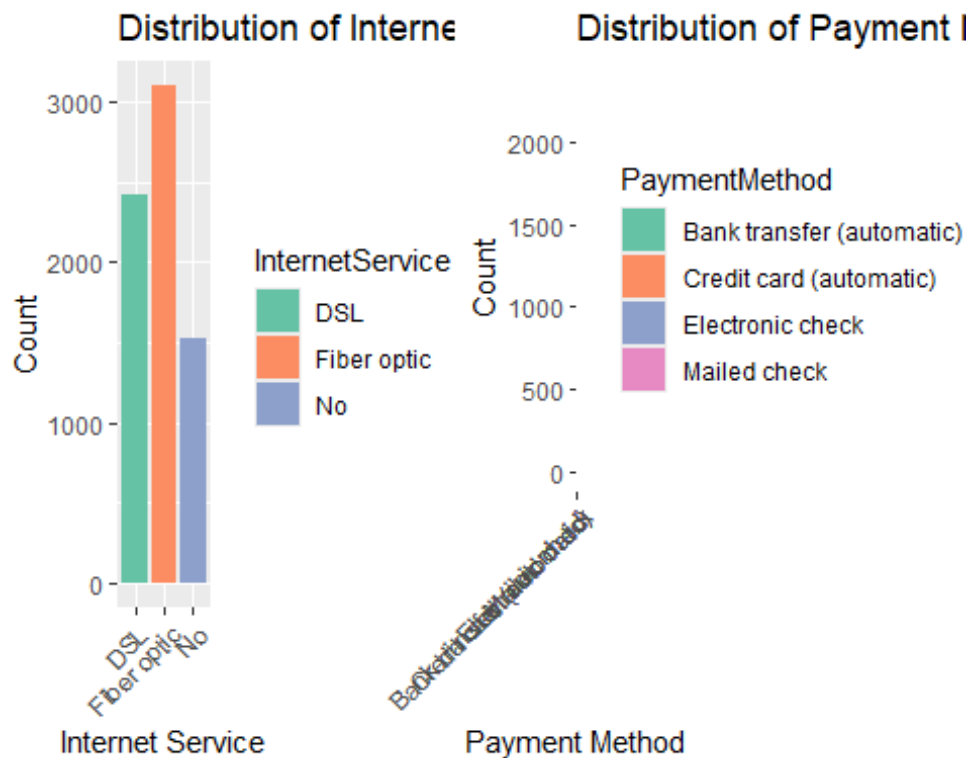
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
## combine

# Bar graph of InternetService
p1 <- ggplot(churn_df, aes(x = InternetService, fill = InternetService)) +
  geom_bar() + labs(title = "Distribution of Internet Service", x = "Internet
Service",
  y = "Count") + scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
# Bar graph of PaymentMethod
p2 <- ggplot(churn_df, aes(x = PaymentMethod, fill = PaymentMethod)) +
  geom_bar() + labs(title = "Distribution of Payment Method", x = "Payment Method",
    y = "Count") + scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Arrange plots in a grid
grid.arrange(p1, p2, nrow = 1, ncol = 2)
```



We can see that Fiber optic is the top choice among internet users, followed by DSL and a small percentage of users not utilizing any internet service. Electronic Check is by far the most popular payment method, with the other three options lagging behind by almost 25%.

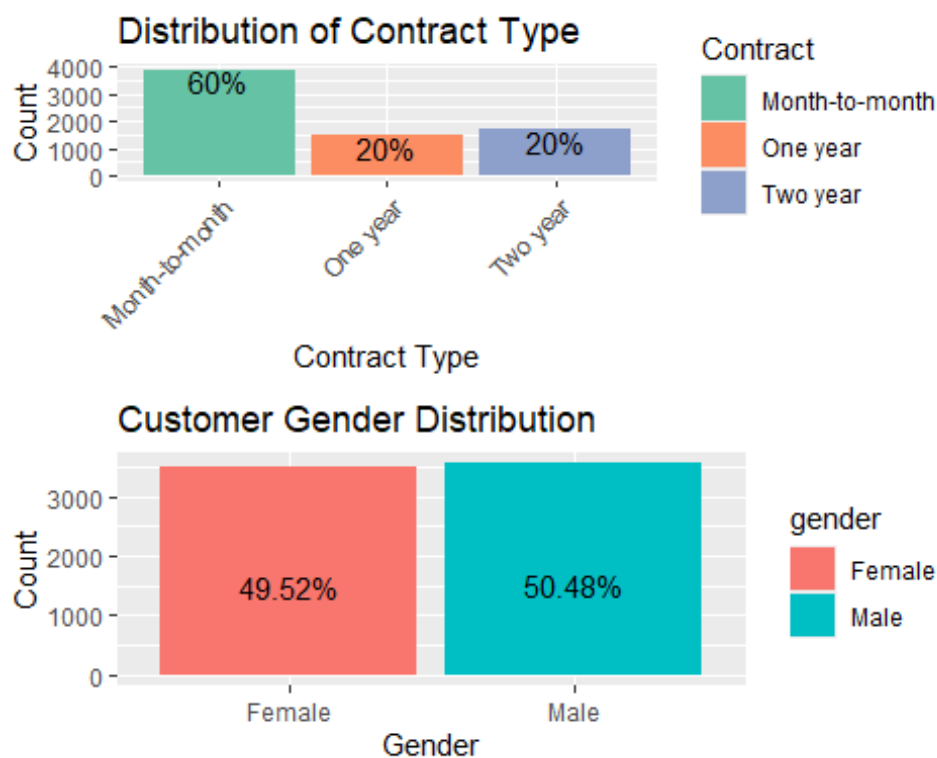
We will create a bar graph of Contract and gender variable

```
# Bar graph of Contract
p3 <- ggplot(churn_df, aes(x = Contract, fill = Contract)) +
  geom_bar() + geom_text(aes(y = ..count.. -400, label =
    paste0(round(prop.table(..count..),
      1)*100, "%")), stat = "count") + labs(title = "Distribution of Contract
    Type",
    x = "Contract Type", y = "Count") + scale_fill_brewer(palette = "Set2") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
# Bar graph of gender
colors <- c("#0072B2", "#E69F00")
p4 <- ggplot(churn_df, aes(x = gender)) + geom_bar(aes(fill = gender)) +
  geom_text(aes(y = ..count.. -500, label = paste0(round(prop.table(..count..), 4)* 100,
  '%')), stat = "count", position = position_stack(vjust = 0.5)) + xlab("Gender") +
  ylab("Count") + ggtitle("Customer Gender Distribution") + theme()

# Arrange plots in a grid
grid.arrange(p3, p4)

## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## [i] Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



The most common contract type is Month-to-Month, followed by Two Year and One Year contracts. The bar graph indicates an almost equal distribution between genders.

Now we will visualize the histogram to show the distribution of numAdminTickets and numTechTickets

```
# histogram of numAdminTickets
p5 <- ggplot(churn_df, aes(x = numAdminTickets)) +
  geom_histogram(binwidth = 1, color = "white", fill = "#FFA07A") +
  geom_text(aes(y = ..count.. -400, label = paste0(round(prop.table(..count..), 4)*100,
  "%")), stat = "count") + labs(title = "Distribution numAdminTickets", x =
  "numAdminTickets", y = "Frequency") + scale_x_continuous(limits = c(-1, 6), breaks =
  seq(0, 5, 1)) +
```

```

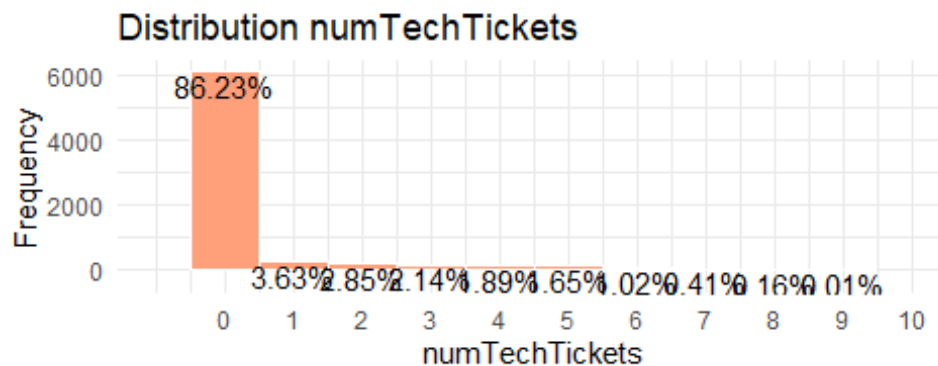
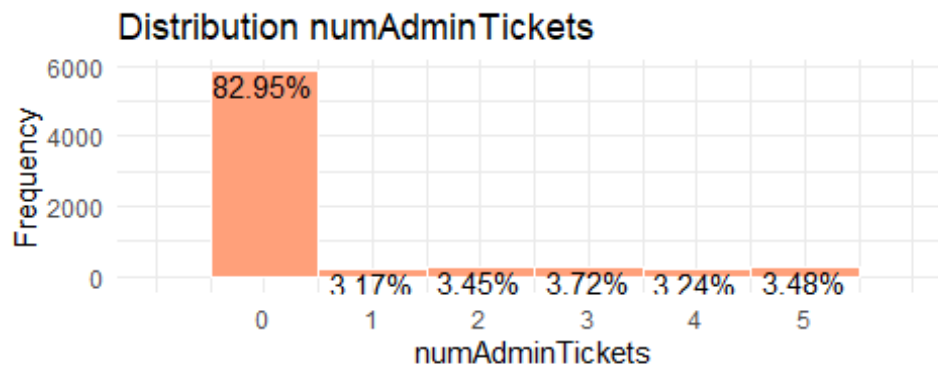
theme_minimal()

# histogram of numTechTickets
p6 <- ggplot(churn_df, aes(x = numTechTickets)) +
  geom_histogram(binwidth = 1, color = "white", fill = "#FFA07A") +
  geom_text(aes(y = ..count.. -400, label = paste0(round(prop.table(..count..), 4)*100,
"%")), stat = "count")+labs(title = "Distribution numTechTickets", x =
"numTechTickets",
y = "Frequency") + scale_x_continuous(limits = c(-1, 10), breaks = seq(0, 10, 1)) +
  theme_minimal()

grid.arrange(p5,p6,ncol = 1)

## Warning: Removed 2 rows containing missing values or values outside the scale
range
## (`geom_bar()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).

```



We will create histogram of Tenure, TotalCharges and MonthlyCharges

```

# Histogram of Tenure
p7 <- ggplot(churn_df, aes(x = tenure)) +
  geom_histogram(binwidth = 5, color = "white", fill = "#FFA07A") + labs(title =
"Distribution of Tenure", x = "Tenure (Months)", y = "Frequency") +
scale_x_continuous(limits = c(0, 80), breaks = seq(0, 80, 5)) + theme_minimal()

# Histogram of TotalCharges

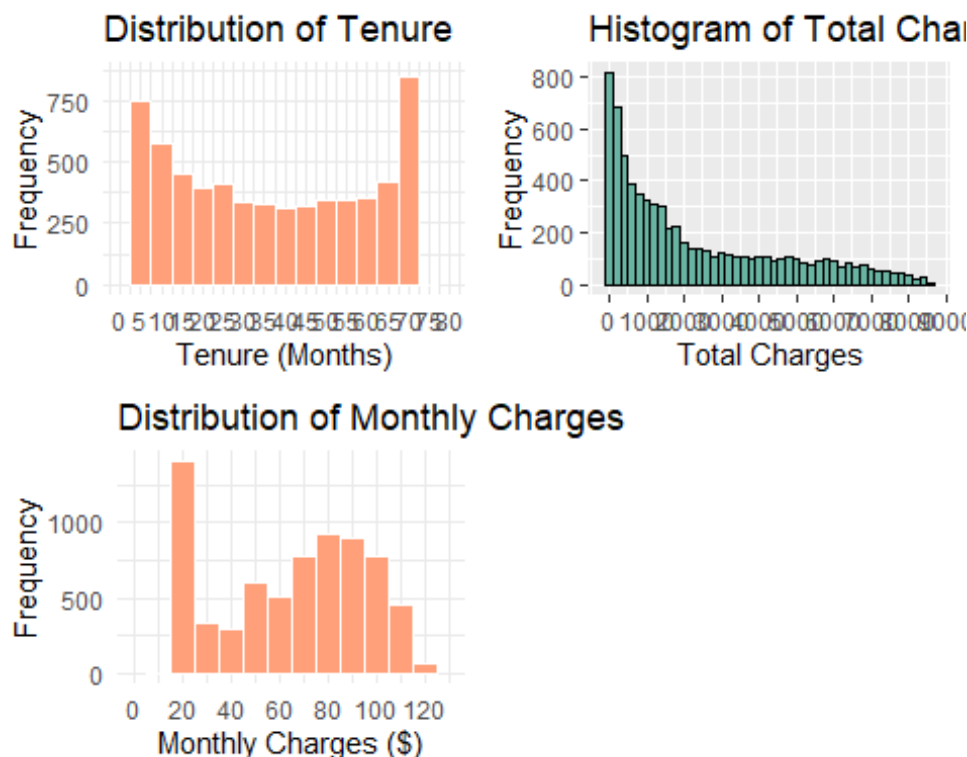
```

```
p8 <- ggplot(churn_df, aes(x = TotalCharges)) +
  geom_histogram(binwidth = 200, fill = "#69b3a2", color = "black") +
  scale_x_continuous(breaks = seq(0, 9000, by = 1000)) + labs(x = "Total Charges", y =
= "Frequency", title = "Histogram of Total Charges")

# Histogram of MonthlyCharges
p9 <- ggplot(churn_df, aes(x = MonthlyCharges)) +
  geom_histogram(binwidth = 10, color = "white", fill = "#FFA07A") +
  labs(title = "Distribution of Monthly Charges", x = "Monthly Charges ($)", y =
"Frequency") + scale_x_continuous(limits = c(0, 130), breaks = seq(0, 150, 20)) +
  theme_minimal()

grid.arrange(p7, p8, p9, ncol = 2)

## Warning: Removed 2 rows containing missing values or values outside the scale
range
## (`geom_bar()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



Maximum frequency (number of customers) is observed in the 0-5 and 65-70 months tenure ranges, whereas the average frequency for intermediate ranges lies between 250 and 400 customers per range. The "TotalCharges" graph depicts the distribution of the charged amount to customers, there is a down trend of decreasing frequency and increasing total charges. The histogram of "Monthly Charges" shows that the majority of customers have monthly charges between 20-40 dollars and a normal distribution around 80 dollars.

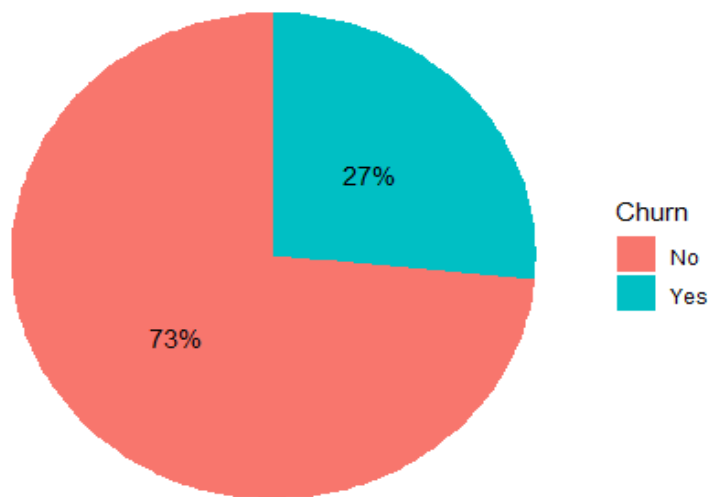
We create a pie chart to show the distribution of churn in the churn_df dataset. It counts how many entries belong to each category ('Churn' or 'No Churn'), converts this count into a dataframe, and then uses ggplot2 to plot the data as a pie chart with percentage labels


```
# Count the occurrences of each churn value
churn_counts <- table(churn_df$Churn)

# Convert churn_counts to a dataframe
churn <- as.data.frame(churn_counts)
names(churn) <- c("Churn", "Count")

# Create the pie chart
ggplot(churn, aes(x = "", y = Count, fill = Churn)) +
  geom_bar(stat = "identity", width = 1) + coord_polar(theta = "y") +
  geom_text(aes(label = scales::percent(Count / sum(Count))),
            position = position_stack(vjust = 0.5)) +
  ggtitle("Churn Distribution") + theme_void()
```

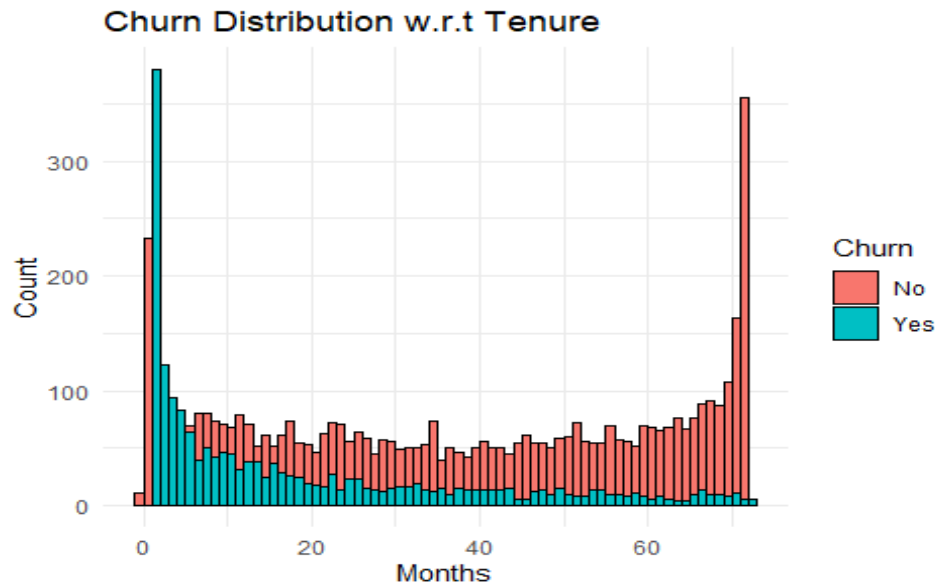
Churn Distribution



Now we will visualize the bar chart to show the distribution of Churn with respect to Tenure

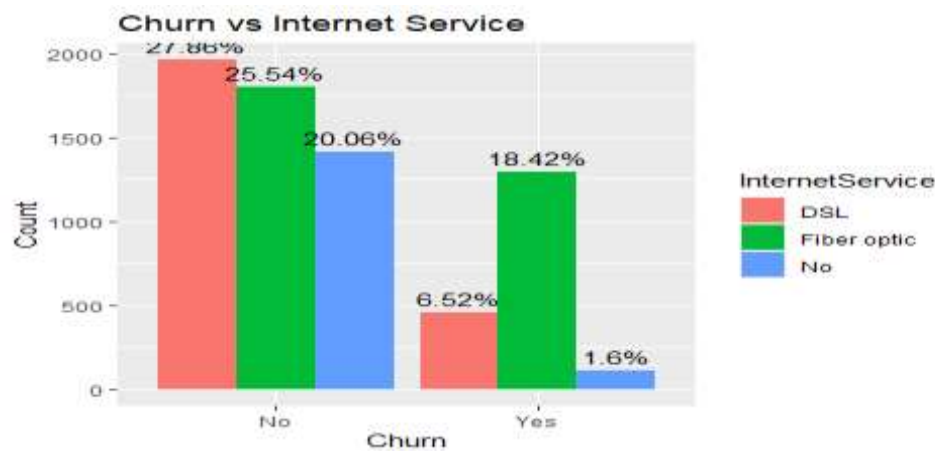
```
ggplot(churn_df, aes(x = tenure, fill = Churn)) +
  geom_bar(position = "dodge", width = 2, colour = "black") +
  labs(title = "Churn Distribution w.r.t Tenure", x = "Months", y = "Count") +
  theme_minimal()

## Warning: `position_dodge()` requires non-overlapping x intervals.
```



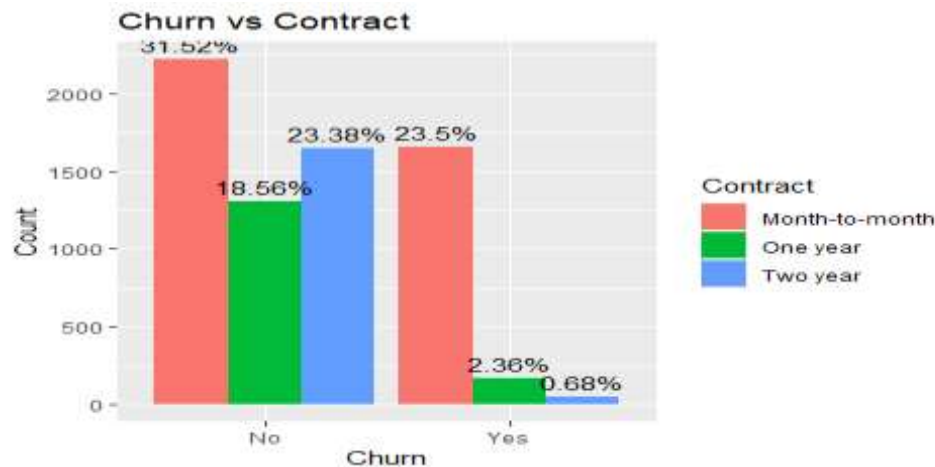
Now we will create a bar plot to show the distribution of churn with respect to Internet Services

```
# bar graph of Internet Service Vs Churn
ggplot(churn_df, aes(x = Churn, fill = InternetService)) + geom_bar(position =
"dodge") + geom_text(aes(y = ..count.., label = paste0(round(prop.table(..count.. /
sum(..count..)), 4) * 100, "%")), stat = "count", position = position_dodge(width =
0.9), vjust = -0.5) +
labs(title = "Churn vs Internet Service", x = "Churn", y = "Count")
```



Now we will create a bar plot to show the distribution of churn with respect to Contract type

```
# Bar Graph of Churn VS Contract type
ggplot(churn_df, aes(x = Churn, fill = Contract)) + geom_bar(position = "dodge") +
geom_text(aes(y = ..count.., label = paste0(round(prop.table(..count.. /
sum(..count..)), 4) * 100, "%")), stat = "count", position = position_dodge(width =
0.9), vjust = -0.5) +
labs(title = "Churn vs Contract", x = "Churn", y = "Count")
```

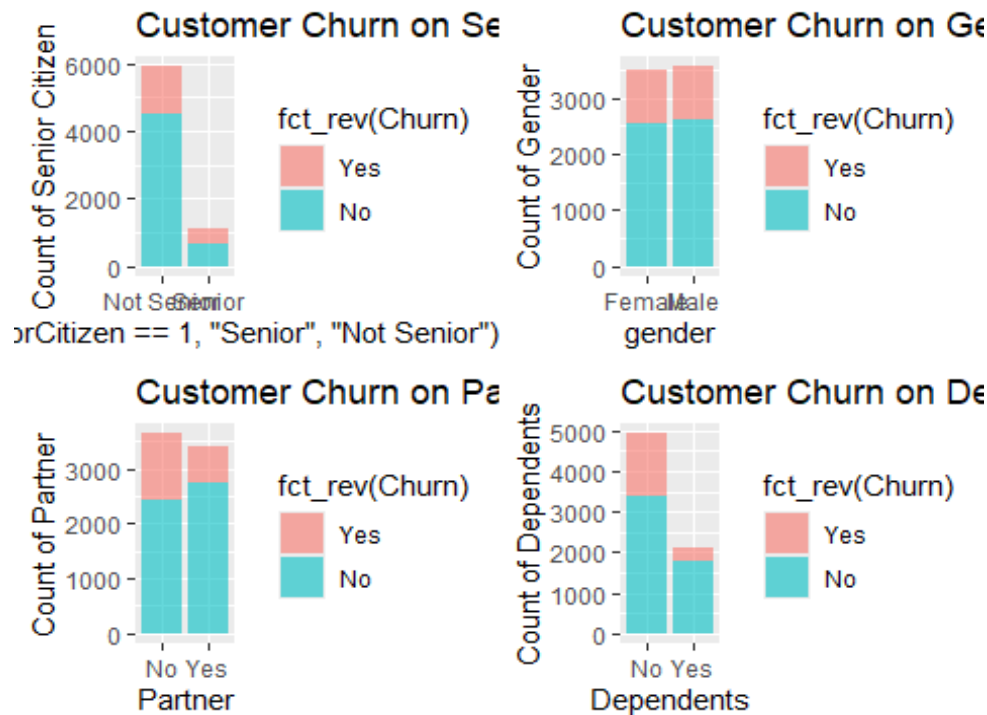


From the bar graph we can see that more customers pay by month-to-month using electronic checks, and these type of customers have the largest percentage of customers churned. Customer under one-year or two-year contracts are much less likely to churn. Customers using automatic payment methods (bank transfer and credit card), as well as mailed check are less likely to churn. Customers using Fiber Optic are most likely to churn.

We will visualize stacked bar chart to show the distribution of churn with respect to Senior Citizen, Gender, Partner and Dependents

```
g1 <- churn_df %>% ggplot(aes(x=ifelse(SeniorCitizen==1, "Senior", "Not Senior"),
fill=fct_rev(Churn)))+ geom_bar(alpha=0.6) + labs(title="Customer Churn on Senior
Citizens", y="Count of Senior Citizen")
g2 <- churn_df %>% ggplot(aes(x=gender, fill=fct_rev(Churn)))+ geom_bar(alpha=0.6) +
labs(title="Customer Churn on Gender", y="Count of Gender")
g3 <- churn_df %>% ggplot(aes(x=Partner, fill=fct_rev(Churn)) + geom_bar(alpha=0.6)
+ labs(title="Customer Churn on Partner", y="Count of Partner")
g4 <- churn_df %>% ggplot(aes(x=Dependents, fill=fct_rev(Churn)))+
geom_bar(alpha=0.6) + labs(title="Customer Churn on Dependents", y="Count of
Dependents")

grid.arrange(g1, g2, g3, g4)
```

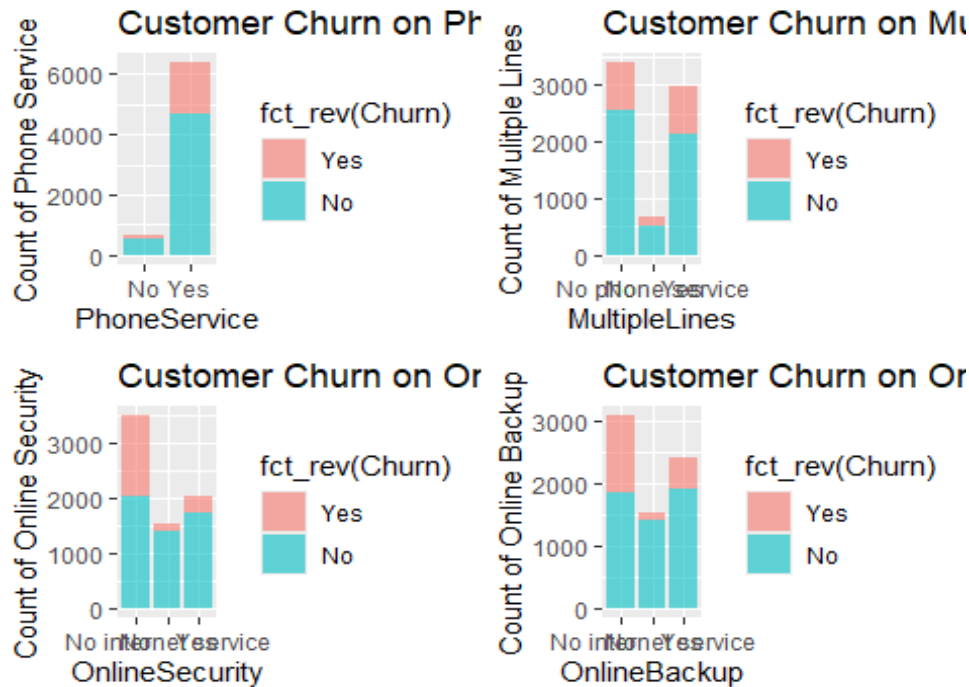


From the histograms above, we can see customer demography, specifically Senior Citizen, Partner, and Dependents, might affect customer churn. We can choose to omit variable Gender in further modeling, but I chose to keep then in the beginning.

We will visualize stacked bar chart to show the distribution of churn with respect to Phone Service, Multiple Lines, Online Security and Online Backup

```
g5 <- churn_df %>% ggplot(aes(x=PhoneService, fill=fct_rev(Churn)))+
  geom_bar(alpha=0.6) + labs(title="Customer Churn on Phone Service", y="Count of Phone Service")
g6 <- churn_df %>% ggplot(aes(x=MultipleLines, fill=fct_rev(Churn)))+
  geom_bar(alpha=0.6) + labs(title="Customer Churn on Multiple Lines", y="Count of Multiple Lines")
g7 <- churn_df %>% ggplot(aes(x=OnlineSecurity, fill=fct_rev(Churn)) +
  geom_bar(alpha=0.6) + labs(title="Customer Churn on Online Security", y="Count of Online Security")
g8 <- churn_df %>% ggplot(aes(x=OnlineBackup, fill=fct_rev(Churn)))+
  geom_bar(alpha=0.6) + labs(title="Customer Churn on Online Backup", y="Count of Online Backup")

grid.arrange(g5, g6, g7, g8, ncol=2)
```



Now we will create a bar plot to show the distribution of churn with respect to StreamingMovies, StreamingTV, TechSupport, DeviceProtection, PaperlessBilling

```
g9 <- ggplot(churn_df, aes(x = Churn, fill = StreamingMovies)) +
  geom_bar(position = "dodge") +
  labs(title = "Churn vs Streaming Movies", x = "Churn", y = "Count")

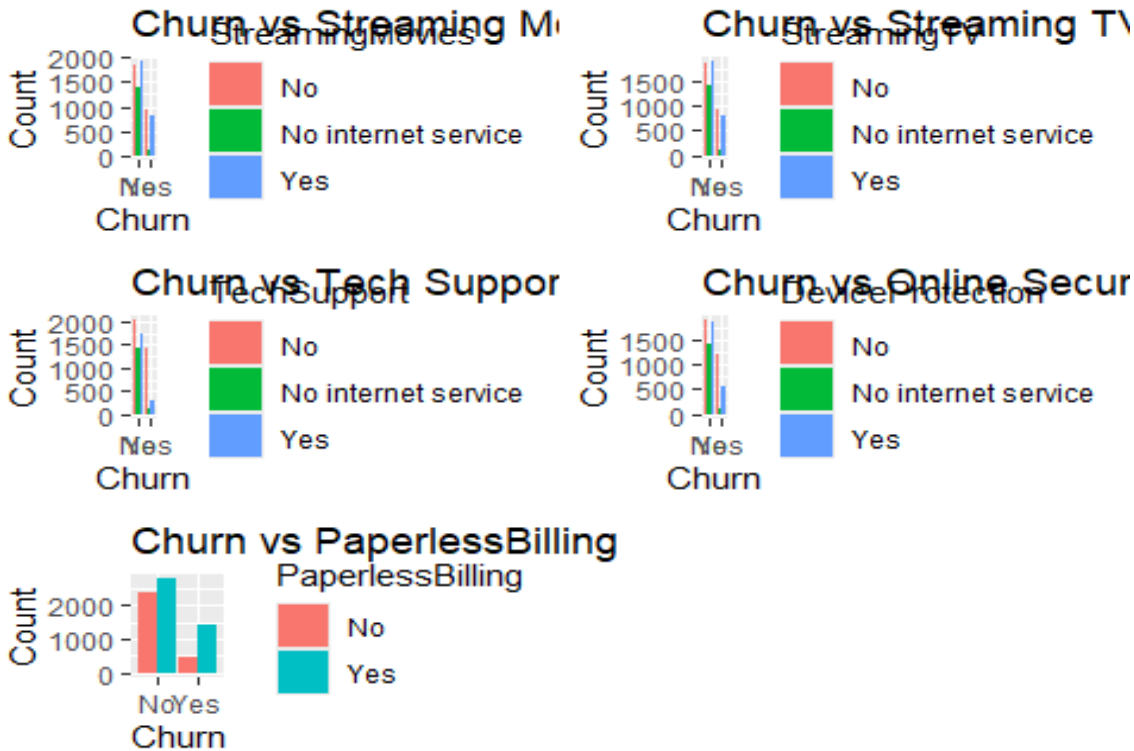
g10 <- ggplot(churn_df, aes(x = Churn, fill = StreamingTV)) +
  geom_bar(position = "dodge") +
  labs(title = "Churn vs Streaming TV", x = "Churn", y = "Count")

g11 <- ggplot(churn_df, aes(x = Churn, fill = TechSupport)) +
  geom_bar(position = "dodge") +
  labs(title = "Churn vs Tech Support", x = "Churn", y = "Count")

g12 <- ggplot(churn_df, aes(x = Churn, fill = DeviceProtection)) +
  geom_bar(position = "dodge") +
  labs(title = "Churn vs Online Security", x = "Churn", y = "Count")

g13 <- ggplot(churn_df, aes(x = Churn, fill = PaperlessBilling)) +
  geom_bar(position = "dodge") +
  labs(title = "Churn vs PaperlessBilling", x = "Churn", y = "Count")

grid.arrange(g9,g10,g11,g12,g13,ncol = 2, nrow =3)
```



From the plot we can see, Multiple Lines, Streaming TV, Streaming Movies have very similar patterns. Online Security and TechSupport have very similar patterns. Online

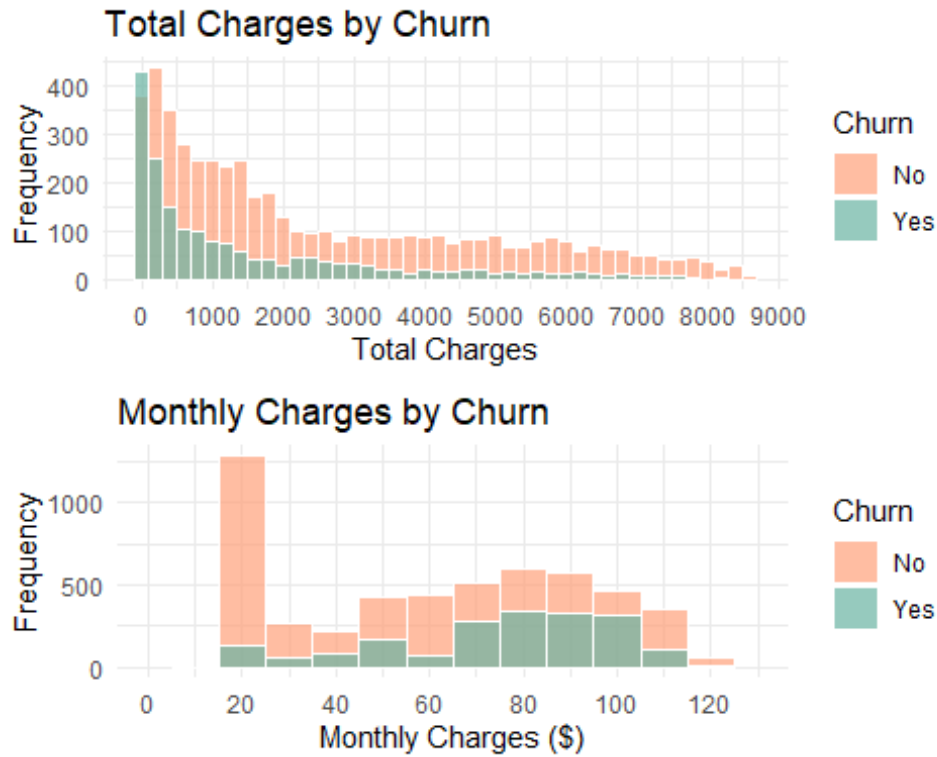
Now we will create a stacked column chart to show the distribution of churn with respect to TotalCharges and MonthlyCharges

```
# TotalCharges Vs Churn :
g14 <- ggplot(churn_df, aes(x = TotalCharges, fill = Churn)) +
  geom_histogram(binwidth = 200, color = "white", position = "identity", alpha = 0.7)
+
  scale_x_continuous(breaks = seq(0, 9000, by = 1000)) +
  labs(x = "Total Charges", y = "Frequency", title = "Total Charges by Churn") +
  theme_minimal() + scale_fill_manual(values = c("No" = "#FFA07A", "Yes" = "#69b3a2"))

# MonthlyCharges Vs Churn :
g15 <- ggplot(churn_df, aes(x = MonthlyCharges, fill = Churn)) +
  geom_histogram(binwidth = 10, color = "white", position = "identity", alpha = 0.7)
+
  labs(title = "Monthly Charges by Churn", x = "Monthly Charges ($)", y =
"Frequency") + scale_x_continuous(limits = c(0, 130), breaks = seq(0, 150, 20)) +
  theme_minimal() + scale_fill_manual(values = c("No" = "#FFA07A", "Yes" =
"#69b3a2"))

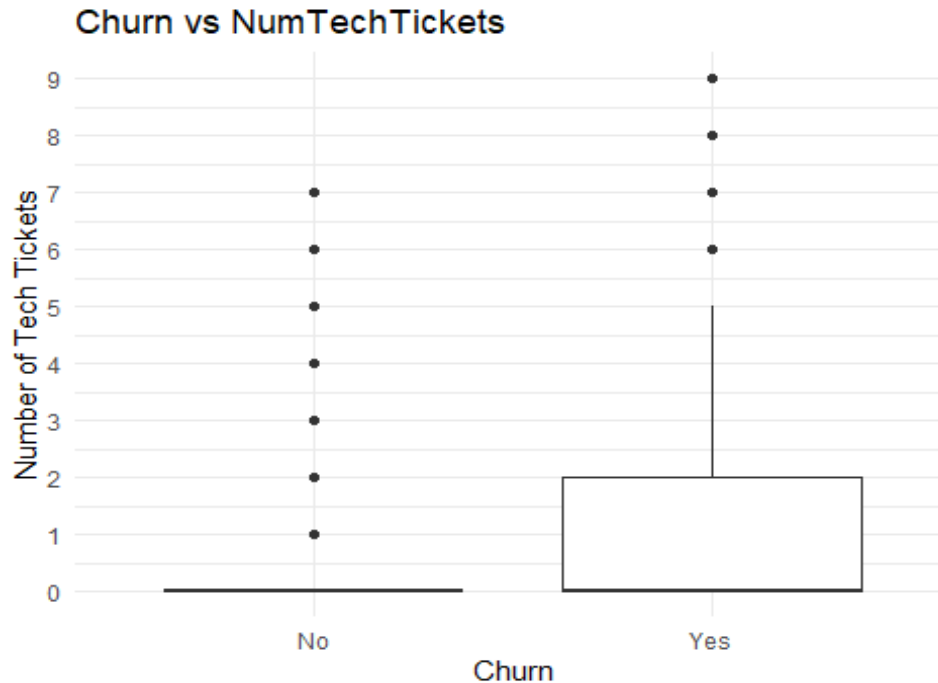
grid.arrange(g14,g15, ncol = 1)

## Warning: Removed 4 rows containing missing values or values outside the scale
range
## (`geom_bar()`).
```



We will create a boxplot to show the distribution of churn with respect to numTechTickets

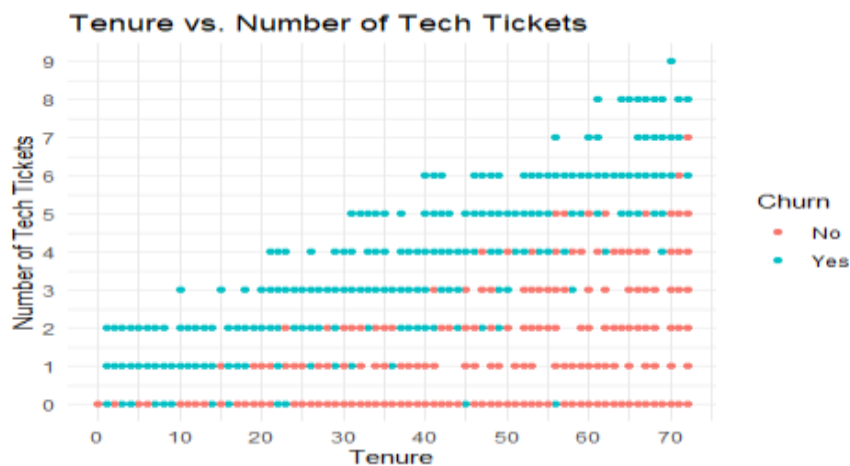
```
# Churn vs numTechTickets
ggplot(churn_df, aes(x = Churn, y = numTechTickets)) +
  geom_boxplot() + scale_y_continuous(limits = c(0,9), breaks = seq(0, 9, 1)) +
  theme_minimal() +
  labs(title = "Churn vs NumTechTickets", x = "Churn", y = "Number of Tech Tickets")
```



The boxplot suggests that customers who churned generally had more tech tickets than those who did not churn

We will create a dot plot to show the distribution of churn with respect to tenure vs numTechTickets

```
# numTechtickets vs Churn
ggplot(churn_df, aes(x = tenure, y = numTechTickets, color = Churn)) +
  geom_point() + labs(title = "Tenure vs. Number of Tech Tickets", x = "Tenure",
    y = "Number of Tech Tickets") + scale_x_continuous(breaks = seq(0, 100, 10)) +
  scale_y_continuous(breaks = seq(0, 10, 1)) + theme_minimal()
```



Customers with a short tenure who raise fewer tech tickets are more likely to churn. Similarly, customers with a longer tenure who raise more than three tech tickets are also more likely to churn. Customers with a tenure greater than 30 and who raise fewer than three tech tickets are much less likely to churn.

Model Training, Testing and Validation

Split data into 70%-30% train and test datasets

```
set.seed(123)
trainIndex <- createDataPartition(churn_df$Churn, p = .7, list=FALSE, times=1)
train <- churn_df[trainIndex,]
test <- churn_df[-trainIndex,]
dim(train)

## [1] 4931  22

dim(test)

## [1] 2112  22
```

Random Forest

Fit the parameters of the model to the training data

```
library(randomForest)

## randomForest 4.7-1.1

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:gridExtra':
##
##   combine

## The following object is masked from 'package:ggplot2':
##
##   margin

## The following object is masked from 'package:dplyr':
##
##   combine

Model1 <- randomForest(as.factor(Churn) ~ ., data = train, mtry = 4, ntree = 400,
                       importance= TRUE)

Model1

##
## Call:
## randomForest(formula = as.factor(Churn) ~ ., data = train, mtry = 4,      ntree =
400, importance = TRUE)
##              Type of random forest: classification
##              Number of trees: 400
```

```
## No. of variables tried at each split: 4
##
##          OOB estimate of  error rate: 14.6%
## Confusion matrix:
##          No Yes class.error
## No   3331 291  0.08034235
## Yes   429 880  0.32773109
```

```
importance(Model1)
```

	No	Yes	MeanDecreaseAccuracy	MeanDecreaseGini
gender	3.0881767	-1.4100585	1.3028513	29.623337
SeniorCitizen	7.6219669	-0.1770552	6.4619159	24.594259
Partner	3.5467946	2.2408332	4.8332660	27.901166
Dependents	-0.4299897	4.1044150	2.6818834	21.447556
tenure	24.8878763	31.9214982	43.9376909	243.970930
PhoneService	1.6559123	7.1494812	6.4414996	6.429166
MultipleLines	8.0888225	7.0879697	12.6861334	32.718231
InternetService	14.4288515	16.2051420	20.1848716	55.232360
OnlineSecurity	3.3510126	27.5128411	15.5777271	75.466882
OnlineBackup	4.2385853	15.7304580	11.5827858	39.918204
DeviceProtection	5.5102151	7.8830348	8.5766449	36.702417
TechSupport	7.5181760	18.3634886	15.0261498	58.872011
StreamingTV	6.0330109	6.8978294	8.6583945	29.743179
StreamingMovies	7.0928826	6.3779949	9.2693950	28.680376
Contract	-1.2913496	26.9405703	26.6519150	125.837849
PaperlessBilling	3.5233828	6.9727433	7.5517001	28.068266
PaymentMethod	4.6137311	8.5056091	10.1214038	84.323600
MonthlyCharges	18.1136842	23.5967051	33.3559483	202.677365
TotalCharges	22.7915521	27.4812703	36.8951792	251.080784
numAdminTickets	1.2499002	-0.1877305	0.8716861	37.400007
numTechTickets	52.3472717	109.3577134	107.2913113	307.600815

We will check top 10 most important features from all the variables in the churn dataset

```
feature_importance <- importance(Model1)
top_10_features <- sort(feature_importance[, "MeanDecreaseAccuracy"],
                        decreasing = TRUE, index.return = TRUE)$ix[1:10]
feature_names <- rownames(feature_importance)[top_10_features]
top_10_importance_values <- feature_importance[top_10_features,
        c("No", "Yes", "MeanDecreaseAccuracy", "MeanDecreaseGini")]
data.frame(Feature = feature_names, top_10_importance_values)
```

	Feature	No	Yes	MeanDecreaseAccuracy
numTechTickets	numTechTickets	52.347272	109.35771	107.29131
tenure	tenure	24.887876	31.92150	43.93769
TotalCharges	TotalCharges	22.791552	27.48127	36.89518
MonthlyCharges	MonthlyCharges	18.113684	23.59671	33.35595
Contract	Contract	-1.291350	26.94057	26.65192
InternetService	InternetService	14.428852	16.20514	20.18487
OnlineSecurity	OnlineSecurity	3.351013	27.51284	15.57773
TechSupport	TechSupport	7.518176	18.36349	15.02615
MultipleLines	MultipleLines	8.088823	7.08797	12.68613
OnlineBackup	OnlineBackup	4.238585	15.73046	11.58279

```

##                               MeanDecreaseGini
## numTechTickets                307.60081
## tenure                        243.97093
## TotalCharges                  251.08078
## MonthlyCharges                202.67736
## Contract                      125.83785
## InternetService               55.23236
## OnlineSecurity                75.46688
## TechSupport                   58.87201
## MultipleLines                 32.71823
## OnlineBackup                  39.91820

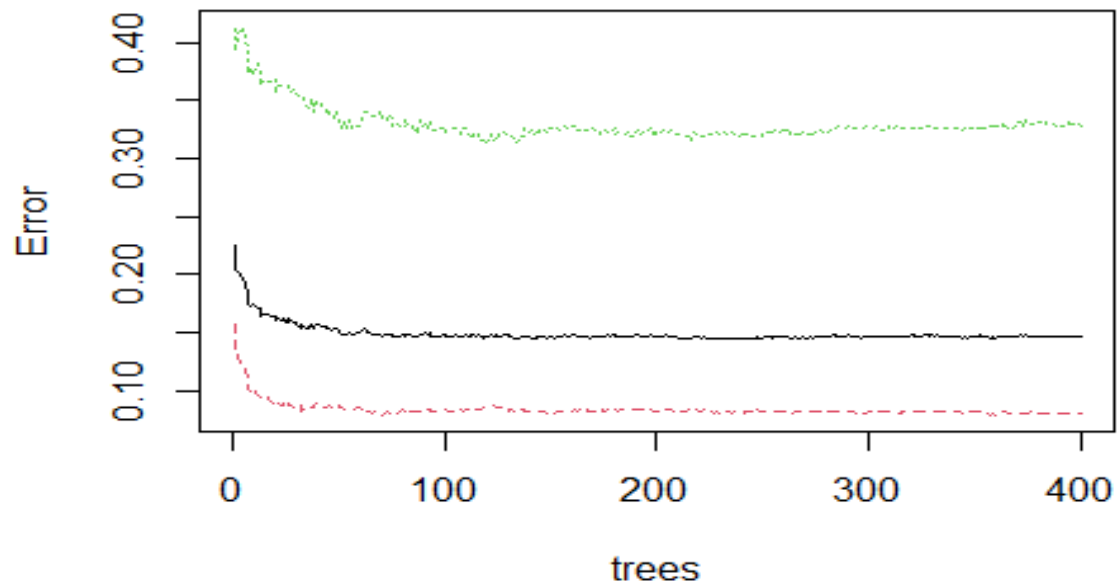
prediction <- predict(Model1,test)
confusion_matrix <- confusionMatrix(as.factor(prediction), as.factor(test$Churn))
confusion_matrix

## Confusion Matrix and Statistics
##
##               Reference
## Prediction    No  Yes
##      No  1433  175
##      Yes   119  385
##
##               Accuracy : 0.8608
##               95% CI : (0.8453, 0.8753)
##      No Information Rate : 0.7348
##      P-Value [Acc > NIR] : < 2.2e-16
##
##               Kappa : 0.631
##
##      Mcnemar's Test P-Value : 0.001338
##
##               Sensitivity : 0.9233
##               Specificity : 0.6875
##               Pos Pred Value : 0.8912
##               Neg Pred Value : 0.7639
##               Prevalence : 0.7348
##               Detection Rate : 0.6785
##      Detection Prevalence : 0.7614
##               Balanced Accuracy : 0.8054
##
##      'Positive' Class : No
##

# Plotting model
plot(Model1)

```

Model1

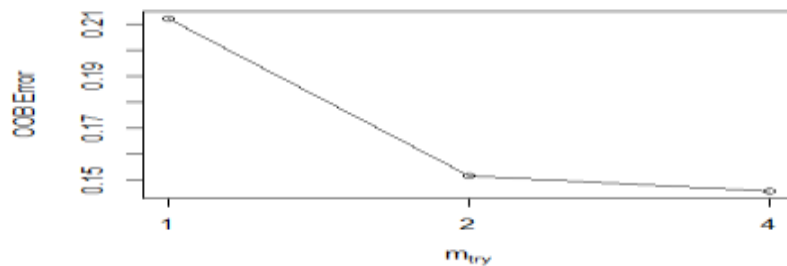


Model Validation

Hyper parameter Tuning

```
set.seed(1)
res <- tuneRF(x = train %>% dplyr::select(-Churn), y = train$Churn, mtryStart=2,
ntreeTry = 500)

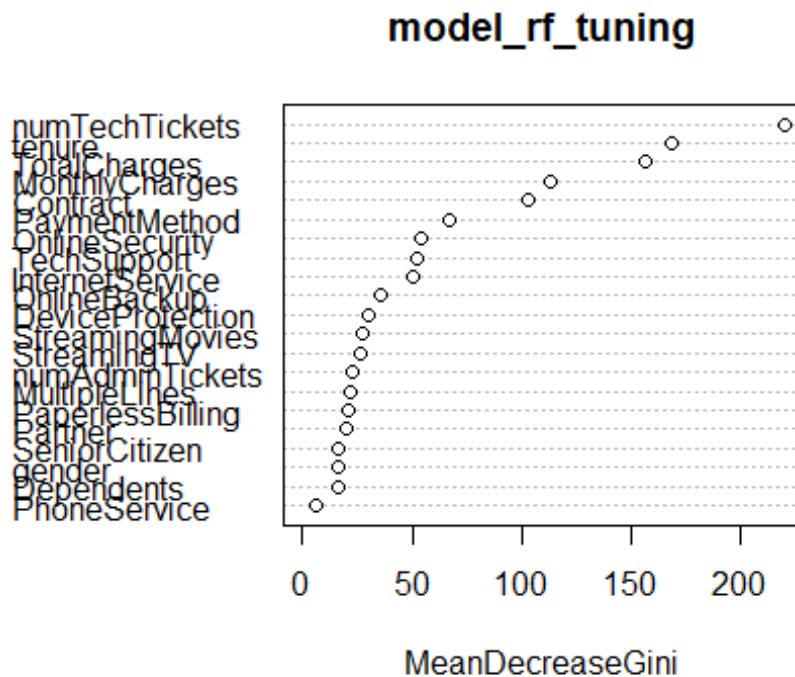
## mtry = 2  OOB error = 15.19%
## Searching left ...
## mtry = 1    OOB error = 21.19%
## -0.3951936 0.05
## Searching right ...
## mtry = 4    OOB error = 14.6%
## 0.03871829 0.05
```



We can see that when mtry=2, the model gives the lowest OOB error. Thus I used mtry=2 in tuned random forest.

Mean decrease GINI

```
set.seed(1)
model_rf_tuning <- randomForest(Churn~., train, ntree=500, mtry=2)
varImpPlot(model_rf_tuning)
```



Tuned Random Forest Validation

We use only the variables with highest mean decrease gini Accuracy of tuned Random Forest:

```
model_rf_tuned <-
randomForest(Churn~tenure+MonthlyCharges+Contract+PaymentMethod+InternetService+OnlineSecurity+PaperlessBilling+TechSupport, train, ntree=500, mtry=2)
test$rf_vote_tuned <- predict(model_rf_tuned, test, type="class")
mean(test$rf_vote_tuned==test$Churn)

## [1] 0.8035038
```

Confusion matrix of tuned Random Forest:

```
rf_tuned_cm <- table(test$rf_vote_tuned, test$Churn)
rf_tuned_cm
```

```
##  
##           No  Yes  
##    No 1401  264  
##    Yes  151  296  
  
rf_tuned_recall <- rf_tuned_cm[2,2]/(rf_tuned_cm[2,2]+rf_tuned_cm[1,2])  
rf_tuned_recall  
## [1] 0.5285714
```